VO_2 Max / Cardiorespiratory Fitness — Engineering Implementation Plan

Extension of the afib-face-scan platform (AvatarX/code/afib) — v0.1 draft for engineering

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1. Scope, Verdict Constraints, and Non-Goals

This plan translates the feasibility report *AvatarX_Contactless_VO2Max_CRF_FaceScan_Feasibility.pdf* (research/, Aug 22, 2026) into engineering work on the existing `AvatarX/code/afib` platform. The science verdict that constrains everything built here:

- **No exact VO_2 max is ever displayed.** Outputs are (1) MEASURED recovery physiology (end-exercise HR proxy, $HRR_{30/60/120}$, recovery slope, resting HR/RR), (2) an INFERRED fitness *category* with uncertainty (gated, off by default), and (3) a within-person trend.
- **A standardized activity is mandatory for any fitness output.** Resting-only scans produce vitals, never fitness.
- **HR is never measured during movement.** The camera's job during the activity is workload verification (rep/cadence counting), not physiology.
- Fitness-category rendering is blocked by pre-registered promotion gates (new `configs/gates.yaml` §V), exactly like the v0.3 ECG-reconstruction track.

Non-goals (standing, mirrors spec §B.7): in-motion rPPG HR; PRV/"HRV"-based fitness features; rBCG stroke-volume proxies; pulse-morphology fitness features; weight inferred from the face; any mL/kg/min point estimate on any surface; any disease/functional-capacity claim.

2. What the Platform Already Provides (Reuse Unchanged)

The production path (spec §B.3) `frames → face track → 4 ROIs → per-ROI rPPG → per-ROI beat detect → consensus fusion → BeatLattice (L3)` is reused as-is for the two stationary phases (rest, recovery). Specifically:

Existing module	Reused for VO_2 work	Change
`capture/` (video_reader, face_tracking, ingest)	Rest + recovery phases	None
`preprocessing/roi.py` (4 facial ROIs)	Same	None
`rppg/pos.py`, `rppg/chrom.py`, `rppg/signal_quality.py`	Same; anti-periodicity SQI stays a permanent gate	None
`beats/` (detector, confidence, ibi, lattice)	Beat times feed the new HR(t) tracker	None to detector; see §3.2 for clean-run caveat
`datasets/` (schema, splits, synchronization, campaigns, registry)	Extended with session/challenge/CPET blocks	Additive schema rev
PRBS LED-marker sync (`datasets/synchronization.py`)	Reference-ECG alignment in recovery-phase validation rigs	None
`inference/` (pipeline, decision_logic, evidence, confidence_stars)	Same gates-first pattern; stars reused for recovery-quality grade	Additive
`heads/base.py` registry (invariant 14: extension = new head)	Three new heads (§4)	Additive
`models/registry.py` promote-refusal on red gates	Fitness head promotion	Point §V gates file
`evaluation/`, falsification lab, `scripts/why_not_ready.py`	New fitness metrics + falsification runs	Additive
`app/` (scan_engine, server, report)	Extended to a three-phase session	§6
Participant-only stable-hash splits + leakage guards	All fitness training/eval	None — mandatory

3. Architecture Delta: The Three-Phase Session

3.1 Session protocol layer (new: `protocol/`)

The single 30-s `ScanSession` becomes one phase of a `ProtocolSession` state machine:

```
SAFETY_SCREEN → REST_SCAN (60 s, existing scan path)
  → GUIDED_ACTIVITY (60–180 s; audio cadence; pose-based rep/cadence
     verification; NO physiology)
  → TRANSITION (target ≤ 5 s to stillness; hard timeout 10 s)
  → RECOVERY_SCAN (120–180 s, existing scan path + HR(t) tracker)
  → QUALITY GATE (ACCEPT / REPEAT_SCAN / NO_RESULT, per phase)
  → heads: recovery → fitness (gated) → trend
```

New files: `protocol/session.py` (state machine, timers, per-phase manifests), `protocol/challenges.py` (protocol definitions: `sts_1min` default, `step_3min`, `march_2min`; each declares cadence, duration, workload formula inputs, contraindication list), `protocol/safety.py` (PAR-Q+-style screen; stop rules; renders only sanctioned text). Each phase writes its own recording manifest so `cli.py validate` gates apply per phase; the transition-time and cadence-compliance results are recorded in provenance and feed the quality gate.

3.2 Recovery HR tracking (new: `features/recovery.py`)

The rhythm pipeline's `clean_runs → compute_rhythm_features_from_runs` assumes quasi-stationary IBI and is wrong for recovery, where HR falls 20–40 bpm in the first minute. Do not reuse the rhythm feature path. New estimator consuming the BeatLattice directly: short-window (5–10 s, 50% overlap) trimmed-median HR with a physiological slew bound ($|dHR/dt| \leq 3$ bpm/s), monotone-trend robust fit, per-window confidence from existing beat confidence + SQI. Outputs: `hr_end_proxy` (back-extrapolation of the 0–15 s fit to $t = 0$), `hrr30/60/120` (deltas vs `hr_end_proxy`), `recovery_slope`, per-window quality vector. Exponential τ is computed for research telemetry only (feasibility report: field CV ~25–35% — not a product metric).

3.3 Activity verification (new: `activity/`)

`activity/pose_cadence.py`: rep counting + cadence estimate for STS/step/march from pose landmarks (MediaPipe pose landmarker via the existing optional-model pattern, `AVATARX_MEDIAPIPE_MODEL`-style env; documented fallback: vertical motion-energy periodicity from the existing tracker — recorded in provenance like the face-tracker chain). Contract: reps within ±10% of prescription and cadence deviation ≤ 10% → compliant; 10–20% → REPEAT_SCAN; > 20% or transition > 10 s → NO_RESULT for fitness outputs (vitals still render). `activity/workload.py`: workload context from user-entered measured weight/height/age/sex/protocol (never inferred from the face).

3.4 Respiratory rate (new: `rppg/respiration.py`, rest phase only)

Chest/shoulder-ROI motion RR at rest (validated range 8–25/min, mirroring QME clearances). Recovery-phase RR is computed but tagged research-only until validated — no rendering.

4. Heads, Schema, and Measurement Classes

Three new heads via the existing registry (`heads/base.py`); none can bypass L2 gates.

Head	Class	Value payload	Rendering rule
head_recovery v1	MEASURED	hr_rest, rr_rest, hr_end_proxy, hrr30/60/120, recovery_slope, protocol_id, workload_context, per-window quality	Renderable once §V1–V2 measurement gates are green; wellness wording from sanctioned tables only
head_fitness v1	INFERRED FITNESS (new enum member)	category (below / typical / above, for age + sex), uncertainty band, inputs_used, baseline-ladder audit	**Off by default; registry.promote refuses while any §V gate is red**; never emits a number in mL/kg/min
head_trend v1	INFERRED FITNESS	direction, magnitude class, n_sessions, window	Requires ≥ 3 accepted sessions per window; local on-device trend store (versioned, exportable)

Schema/invariant updates (additive, single rev): extend the "app may render MEASURED and INFERRED_RHYTHM only" invariant to include INFERRED_FITNESS *only when the §V gate file is green and signed*; add session blocks `challenge{protocol_id, cadence_prescribed/achieved, reps, transition_s}`, `participant_context{age, sex, measured_weight_kg, height_cm, meds{beta_blocker, ccb, ivabradine, stimulant, thyroid}, activity_ipaq}`, and label block `cpet{vo2peak_mlkgmin, modality, protocol, rer_peak, hr_peak, effort_criteria[], avg_window_s, cart, lab, test_date}`. Beta-blocker/rate-limiting medication answers route `head_fitness` to trend-only output (hard rule from the feasibility report).

5. Pre-Registered Promotion Gates (`configs/gates.yaml` §V)

Same shape as §G (v0.3): numeric thresholds are provisional defaults, REQUIRES_CLINICAL_SIGNOFF, loosening requires a signed spec changelog entry; facial-domain, participant- and session-disjoint, production-path evidence only (public finger-PPG/synthetic sets exercise machinery, never open a gate).

- **V1 — recovery-HR fidelity:** HRR_{60} 95% LoA ≤ ±5 bpm vs reference ECG in ACCEPTED scans, every powered Monk band; per-window HR RMSE reported by recovery time. (Kill: > ±8 bpm in any band after engineering → that band gets NO_RESULT.)
- **V2 — protocol repeatability:** HRR_{60} test-retest typical error ≤ 4 bpm; ≥ 85% accepted-scan rate, ages 20–75; no powered band's abstention > 1.5× best and > +10 pts (the platform's existing no-read parity rule applied to the recovery scan).
- **V3 — incremental value (the pivotal gate):** demographics + activity + recovery physiology must beat demographics + activity by ΔSEE ≥ 0.5 mL/kg/min AND category accuracy ≥ +5 pts (CI excluding zero), and must beat the **static-face-image negative control**, on participant-disjoint held-out CPET data. Fail → `head_fitness` is never promoted; product ships recovery metrics only.
- **V4 — subgroup honesty:** beta-blocker bias ≤ 1 MET or medicated users are trend-only (enforced in code, not labeling); per-subgroup (age, sex, BMI, Monk, meds) bias/LoA tables required.
- **V5 — trend validity (for `head_trend` beyond direction-only):** training-response study change-score r ≥ 0.6 / AUROC ≥ 0.75 for a ≥ 1-MET change.

6. Evaluation, Data, and App Work

Evaluation (`evaluation/fitness_metrics.py` + harness): SEE/MAE/bias/Bland–Altman LoA (mL/kg/min), category accuracy/AUROC/calibration, and the mandatory baseline ladder computed in the same run: age; age+sex; +BMI; resting-HR-only; activity-only; demographics+activity; **static face image**; resting physiology; recovery physiology; full — identical participant-level splits, reported side by side in `evaluation_report.{json,md}`. Extend `cli.py evaluate` to accept `<id>.cpet.json` labels; excluded recordings remain EXCLUDED-with-reasons. Falsification lab additions: shuffled-workload test (features must degrade), demographic-only impostor model, transition-time sensitivity sweep.

Bench data (research only, licenses respected): LGI-PPGI (CC-BY) and, under academic terms, MMPD/SUMS/LADH for post-exercise robustness; Malaga treadmill + NHANES CVX for HRR→VO_2 modeling machinery. Nothing trained on restricted sets ships (same rule as MIMIC in the AFib track). Proprietary Stage 1–2 capture (40–60 participants, ECG + contact-PPG reference, ≥ 25% dark skin tones) uses the existing campaign tooling + PRBS sync rig.

App (`app/`): extend the demo flow with safety screen → rest scan → cadence-coached activity (audio metronome, live rep counter, no physiology UI) → “sit still” countdown → recovery scan → findings report. Consumer capture profile rules unchanged; every gate fails closed; report renders MEASURED recovery metrics with the star grade; fitness category appears only behind the §V flag. Telemetry per session appends protocol compliance, transition time, per-phase SQI, rejection reasons.

7. Milestones (tests first, dependency order)

M	Deliverable	Acceptance
M1 (wk 1–2)	`features/recovery.py` + synthetic recovery-video generator (`scripts/make_synth_recovery.py`: known HR(t) decay curves)	HR(t) MAE ≤ 2 bpm and HRR_{60} error ≤ 3 bpm on synthetic decays incl. dropout/motion injections; property tests pin slew bound and back-extrapolation
M2 (wk 2–4)	`protocol/` session state machine + `activity/` pose-cadence with fallback; per-phase manifests through `cli.py validate`	Headless-browser selftest completes full three-phase flow; cadence counter within ±1 rep on scripted videos; transition timer enforced
M3 (wk 3–5)	Schema rev + `configs/gates.yaml` §V + `evaluation/fitness_metrics.py` + baseline-ladder harness + `cli.py evaluate` CPET support	Leakage audit green; ladder report generated on synthetic cohort; promote-refusal test for red §V
M4 (wk 5–7)	Three heads registered; trend store; decision logic + sanctioned text tables; app three-phase UX behind flag	All heads pass registry/contract tests; fitness head unreachable while §V red; report snapshot tests
M5 (wk 7–9)	Stage-1 capture campaign tooling (reference rig configs, PRBS in recovery), public-set benchmarks, falsification runs, README.md update	Benchmarks reproduce with one script each; `why_not_ready.py` lists §V status truthfully

Engineering effort ≈ 2 engineers × 9 weeks to a research-complete v0.1; §V promotion then depends on the data campaigns (feasibility report, Sections 17 and 23), not on code.

8. Standing Rules

Every number surfaced carries its MeasurementClass; sanctioned text only; NO_RESULT is a result; participant-level splits everywhere; loosening any gate requires a signed changelog; the fitness category is a wellness output — no disease names, no clinical thresholds, no "accuracy" claims — per the January 2026 FDA wellness guidance analysis in the feasibility report (§20).